

1. Describe the cause of food grain deficit in the country with the highlight of functional management of irrigation system.

Answer:

Food grain deficit leads to famine in the country resulting in the death of people or suffering from diseases/health conditions such as malnutrition. Functional management refers to the efficient planning, operation, maintenance, and administration of irrigation systems to ensure reliable water supply and sustainable agricultural production. In FY २०८१/८२, percentage of food crops production in Nepal was ४५.५२% and about १५% of food crops were imported.

The major causes of food grain deficit in the country are listed below:

- i. Dependence on monsoon rainfall and inadequate irrigation.
- ii. Small and fragmented land holdings.
- iii. Low agricultural productivity due to traditional farming methods.
- iv. Poor quality seeds and insufficient use of fertilizers.
- v. Frequent floods, droughts, landslides, and climate change impacts.
- vi. Lack of agricultural mechanization and modern technology.
- vii. Poor storage, transportation, and marketing facilities leading to post-harvest losses.
- viii. Rapid population growth and increasing food demand.
- ix. Conversion of agricultural land to urban and industrial uses.
- x. Inadequate investment in agricultural infrastructure and extension services.

Functional management of irrigation systems helps to prevent food grain deficit, as mentioned in points below:

- i. Proper planning and design of irrigation infrastructure.
- ii. Equitable and timely distribution of irrigation water.
- iii. Regular operation and maintenance of canals and structures.
- iv. Formation and strengthening of Water Users' Associations (WUAs).
- v. Participation of farmers in decision-making and maintenance.
- vi. Water conservation and efficient irrigation methods.
- vii. Collection of irrigation service fees for maintenance.
- viii. Monitoring water use and resolving conflicts among users.
- ix. Adoption of modern irrigation techniques such as sprinkler and drip irrigation.
- x. Capacity building and training for farmers and irrigation managers.

2. Agency managed irrigation system in Nepal is performing less efficiently, what are the main reasons behind this? How can they be improved?

Answer:

- Agency managed irrigation system is the irrigation system constructed and operated by the agency which may be a private agency or government itself. A prominent example of a struggling or failed agency-managed irrigation system (AMIS) in Nepal is the Sikta Irrigation Project, Narayani Irrigation Project etc. Some of the biggest FMIS maintained in Nepal are Rajapur Irrigation System, Rani Jamara Kulariya Irrigation System etc.

The reason AMIS in Nepal is performing less efficiently are listed below:

- i. Weak participation of farmers in system management which creates no sense of ownership
- ii. Poor operation and maintenance of canals and structures.
- iii. Inadequate budget and delayed maintenance.
- iv. Inequitable water distribution, especially at the tail end.
- v. Siltation, leakage, and water losses in canals.

- vi. Lack of trained technical and management personnel.
- vii. Poor coordination among government agencies and users.
- viii. Encroachment and damage to irrigation infrastructure.
- ix. Low water-use efficiency and outdated irrigation methods.
- x. Weak monitoring, accountability, and institutional management.

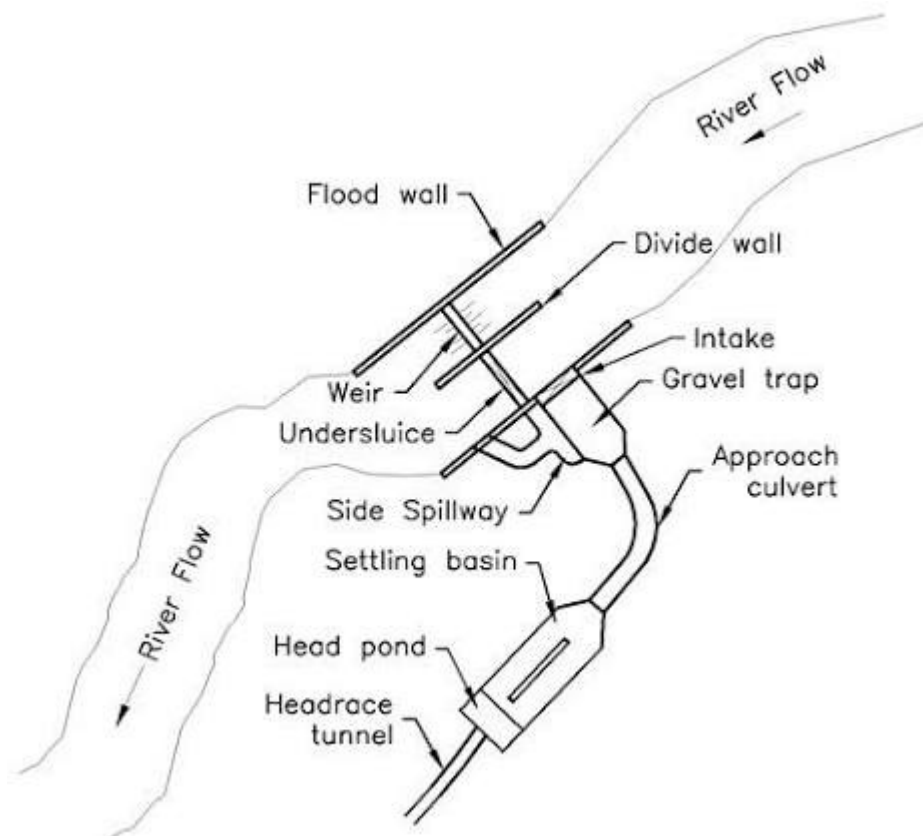
They can be improved by:

- i. Strengthen Water Users' Associations (WUAs) and promote participatory irrigation management.
- ii. Ensure regular operation and maintenance with adequate funding.
- iii. Rehabilitate and modernize old irrigation infrastructure.
- iv. Introduce efficient irrigation methods such as drip and sprinkler systems.
- v. Improve equitable water distribution through proper scheduling and monitoring.
- vi. Provide training and capacity building for farmers and irrigation staff.
- vii. Reduce canal seepage by lining canals where necessary.
- viii. Use modern technologies for water measurement and system monitoring.
- ix. Strengthen coordination among government agencies, local governments, and users.
- x. Establish effective cost recovery and transparent financial management.

3. **Describe the various types of headworks and its main constituent and functions. Which types of headwork is most suitable for the hilly and mountainous region of Nepal? Give reasons.**

Answer:

Headworks are the hydraulic structures constructed at the intake point of a river or reservoir to divert required amount of water into the canal. They help to regulate, control, and safely divert the required quantity of water while preventing sediment and debris from entering the canal.



Types of headworks:

- i. **Temporary (Seasonal) Headworks**
 - Temporary headworks are constructed using local materials such as earth, stones, sandbags, brushwood, and timber.
 - They are built across small rivers or streams.
 - These temporary headworks usually wash away during floods and rebuilt each year.

- They are suitable for small irrigation schemes.
- ii. **Permanent Headworks:**

- Permanent headworks are durable structures made of concrete or masonry. They are classified into:

- a. **Diversion Weir**

- ✓ Diversion weir is a low barrier built across a river to raise the water level.
- ✓ In this type of headwork, water flows over the crest.
- ✓ It is suitable for rivers with small fluctuations in water level.

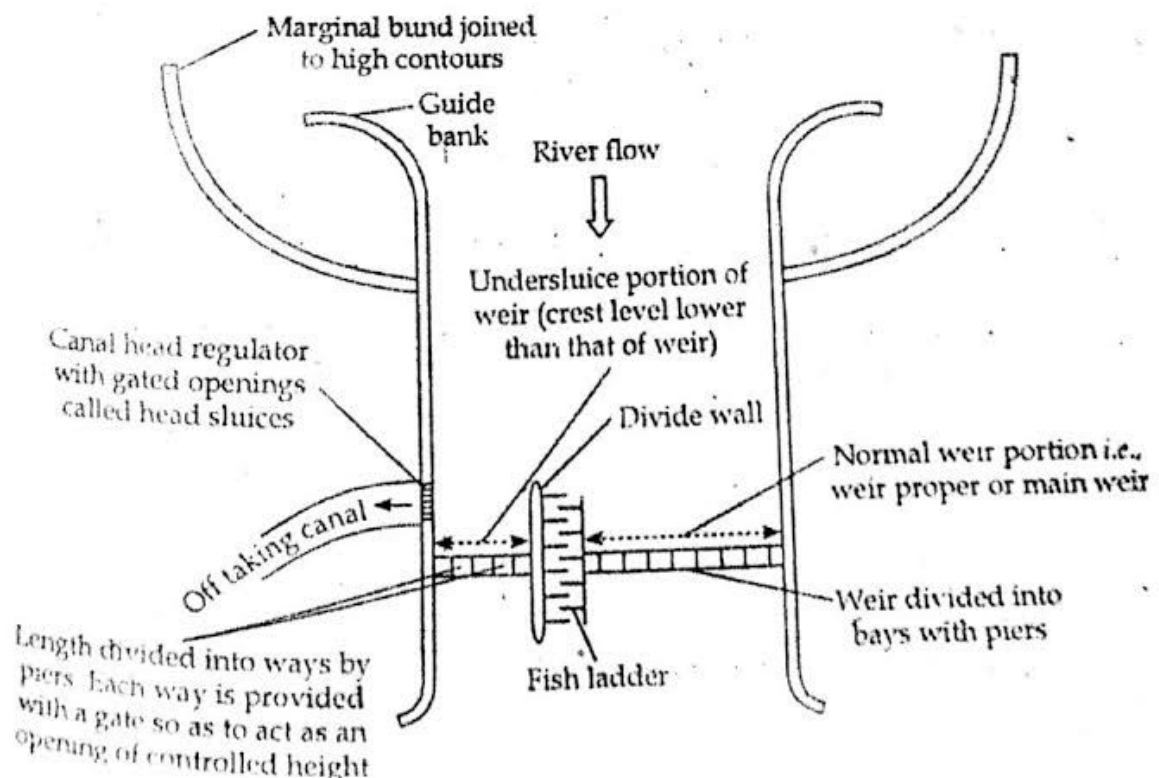
- b. **Diversion Barrage**

- ✓ Barrage is a gated structure with adjustable gates.
- ✓ It regulates river flow and maintains desired upstream water level.
- ✓ It is suitable for large rivers with significant seasonal flow variation.

- c. **Tyrolean (Bottom Rack) Weir/Intake**

- ✓ Water enters through steel bars (bottom rack) placed on the river bed in this type of headwork.
- ✓ Coarse sediment, boulders, and debris continue downstream.
- ✓ It is commonly used in steep mountain streams.

Main constituent of headwork and its functions:



- i. **Weir/barrage:**
The main function of weir/barrage is to raise the water level and divert the flow into canal.
- ii. **Under sluice:**
Undersluice helps to remove deposited sediment and pass flood water.
- iii. **Divide wall:**
It isolates canal head regulator from the main river flow and creates a still pond of water in front of canal head regulator.

- iv. Fish ladder:
It allows migration of fish across the structure.
- v. Canal head regulator:
Canal head regulator regulates and control the amount of water entering into canal.
- vi. Silt excluder:
The main function of silt excluder is to exclude silts from the water source entering the canal.
- vii. Silt ejector:
It removes the sediment entered into the canal and disposes it at the downstream of river.
- viii. River training works:
River training works guide the flow of river and protects the structures and adjoining land.
- ix. Energy dissipation:
Energy dissipator reduces the flow velocity of water downstream and prevent scour.

The most suitable types of headwork for hilly and mountainous region of Nepal is Tyrolean (Bottom Rack) intake because:

- ✓ It effectively excludes boulders, gravel and coarse sediment from entering into the canal.
- ✓ It performs well under flash flood conditions.
- ✓ It is suitable for steep-gradient mountain streams.
- ✓ These intakes are less susceptible to blockage by floating debris.

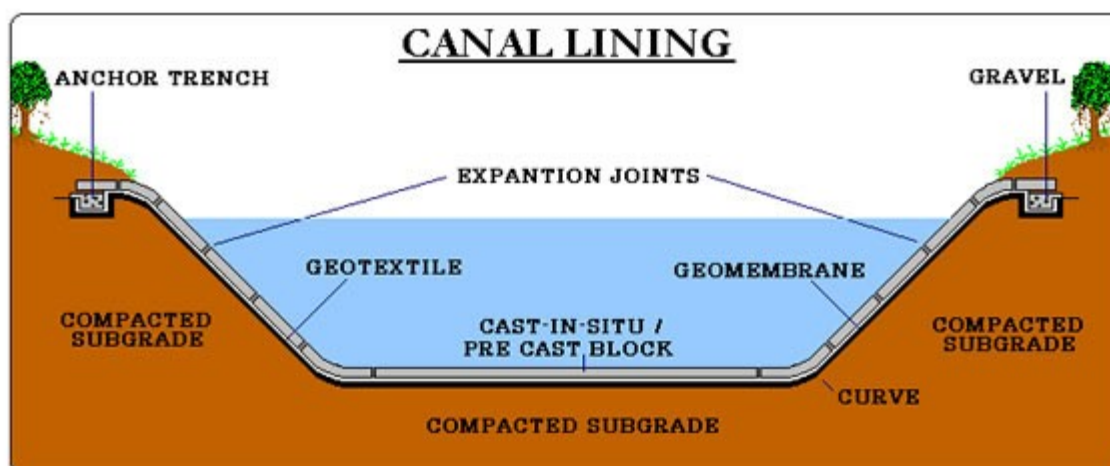
4. **Why canal lining is necessary in light soil? Describe in brief, the various types of lining is used in hills and terai of Nepal?**

Answer:

Canal lining is a method used to improve irrigation efficiency by covering the bed and sides of a canal with impervious materials such as concrete, brick, mud, or steel. Its primary purpose is to reduce seepage losses and enhance the overall performance of the water conveyance system.

Canal lining is necessary because:

- i. It prevents significant water loss due to seepage through the soil, ensuring more water reaches the tail end of the canal.
- ii. The smooth surface of the lining reduces hydraulic resistance, allowing for higher flow velocities and greater discharge capacity.
- iii. It minimizes the growth of weeds, reduces silt deposition, and prevents bank erosion.
- iv. Lining protects the canal bed and banks from scouring caused by high-velocity water flow.



The types of canal lining that is commonly used in hilly region of Nepal:

i. Stone masonry lining:

- ✓ It is constructed using locally available stones with cement mortar.
- ✓ It is suitable for steep slopes and rocky terrain.
- ✓ This type of lining is common in hill irrigation systems.

ii. Dry stone pitching:

- ✓ In dry stone pitching, stones are placed without mortar.
- ✓ They are economical where stones are abundant.
- ✓ Dry stone pitching is used for small canals and temporary protection.

iii. Concrete lining:

- ✓ Concrete lining provides excellent water tightness.
- ✓ They are suitable for canals carrying high discharge.
- ✓ These lining are durable but relatively expensive.

iv. Shotcrete lining:

- ✓ Concrete sprayed under pressure.
- ✓ Used on unstable rocky slopes and difficult hill sections.

The type of lining commonly used in Terai region of Nepal are:

i. Cement concrete lining:

- ✓ Concrete lining provides excellent water tightness.
- ✓ They are suitable for canals carrying high discharge.
- ✓ These lining are durable but relatively expensive.

ii. Brick lining:

- ✓ Brick lining is constructed with burnt clay bricks in cement mortar.
- ✓ They are economical where bricks are readily available.

iii. Geomembrane lining:

- ✓ Geomembrane lining consists of HDPE or PVC sheets covered with soil or concrete.
- ✓ They are highly impermeable and effective in sandy soils.

iv. Compacted clay lining:

- ✓ Compacted clay lining is used where suitable clay is locally available.
- ✓ They are less durable than concrete.

5. **Write short notes on:**

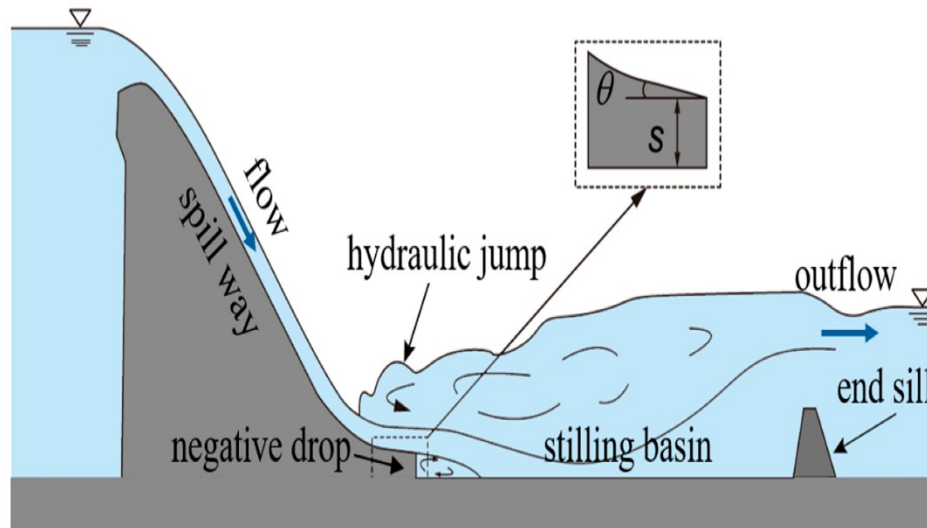
- i. Hydraulic jump:
- ii. Launching Apron
- iii. Water logging
- iv. Surface drainage

Answer:

i. Hydraulic jump:

- When the depth of flow changes rapidly from a low stage to a high stage, there is abrupt rise of water surface. This local phenomenon is known as hydraulic jump.
- It occurs when the flow changes from supercritical to subcritical.
- Hydraulic jump converts kinetic energy of stream flowing rapidly into potential energy.
- It occurs frequently in a channel below a regulating sluice, at toe of spillway and when the channel bed slope suddenly changes from the steeper to flat slope or where the velocity is high.

- Hydraulic jump forms eddy turbulences in water which causes dissipation of energy. It is necessary to be formed in hydraulic structures which is explained in the points below:
 - i. For the dissipation of excess energy of flowing water over spillway and other hydraulic structure.
 - ii. For preventing from possible erosion of downstream floor at structure.
 - iii. For increasing the weight of apron floor of hydraulic structure due to increased flow depth.
 - iv. For mixing chemicals in water by the help of turbulence created.
 - v. For raising the water level in the channel for irrigation.
 - vi. For increasing the discharge through a sluice by holding back tail water.



For a stable and well-balanced hydraulic jump, the Froude number (Fr) before the jump should be greater than 1 i.e., the flow must be supercritical.

The table below shows the Froude number for different type of hydraulic jump.

Froude Number (Fr_1)	Type of Hydraulic Jump
$(Fr = 1)$	No hydraulic jump (critical flow)
$(1 < Fr_1 \leq 1.7)$	Undular jump (weak, unstable)
$(1.7 < Fr_1 \leq 2.5)$	Weak jump
$(2.5 < Fr_1 \leq 4.5)$	Oscillating jump
$(4.5 < Fr_1 \leq 9)$	Stable (steady) hydraulic jump
$(Fr_1 > 9)$	Strong jump

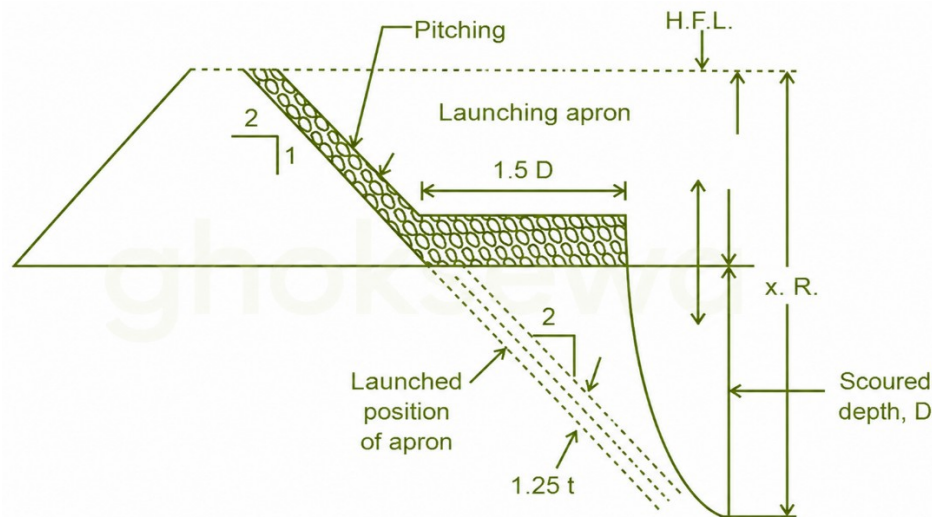
A stable and well-balanced hydraulic jump is generally obtained when:

$$4.5 < Fr_1 \leq 9.0$$

ii. Launching apron:

- A launching apron is a layer of stone, boulders, or concrete blocks placed at the toe of a river training or hydraulic structure (such as a weir, barrage, or river embankment).
- It is designed to launch (settle) automatically into a scour hole formed by flowing water, protecting the structure from erosion.
- It protects the face of guide bund so that during floods the sloping face is not damaged.
- Scour would occur at the toe with consequent undermining and collapse of stone pitching.
- Adequate quantity of stones for launching apron has to be provided to ensure complete protection of whole of the scoured face.
- Generally, scour slope of 2H:1V is assumed.
- In this case, the length of the scoured face will be equal to $\sqrt{5} D$.

- The thickness of apron in launched position is assumed as $1.25t$, where t is the thickness of stone pitching:
- Advantages:
 - i. It is flexible and self-adjusting.
 - ii. It is economical compared to rigid protection.
 - iii. Launching aprons are easy to construct and maintain.
 - iv. It is more effective for deep scour conditions.



Launching Apron Details

iii. Water logging:

Water logging is the condition of saturation of soil such that air phase is restricted and anaerobic conditions prevails. Waterlogging is often accompanied by soil salinity as waterlogged soils prevent leaching of the salts imported by the irrigation water. Due to water logging, groundwater table rises close to the ground surface (generally within 1.5–2.0 m), causing the root zone to remain saturated and reducing soil aeration.

- Water logging is an increasing problem in flat terrains of terai region and low-lying pocket lands in Nepal.
- It is a moderately serious, localized, but recurring seasonal problem.
- It is primarily driven not only by over-irrigation but also by heavy monsoon rainfall combined with unmanaged infrastructure.
- Water-logging is caused by intensive irrigation, seepage from canals/ reservoirs, seepage from adjoining lands, lack of adequate drainage systems, monsoon rainfall and so on.
- Water logging has caused hamper in nitrification which reduces the crop yield and also leads to rise of salt in surface layer and overgrowth of weeds.



iv. Surface drainage:

- The removal of excess rainwater falling on the fields or excess irrigation water applied to the fields by constructing open ditches. Field drains and other structures is known as surface drainage system.
- The land is loped toward the ditches or drains making excess flow of water to remove from the drains.
- Surface drains are needed for removing the storm water and excess farm water for most of the cultivated crops on flat or undulating topography.
- Surface drains can be divided into shallow surface drain and deep surface drains.
 - i. Shallow surface drain:
 - The open drains constructed to remove excess irrigation water collected in the depression on the fields, as well as the storm water are known as shallow surface drains.
 - These drains carry the runoff to the outlet drains.
 - They are trapezoidal in cross section.
 - If designed properly, they should carry the normal storm water from the fields, plus the excess irrigation water.
 - ii. Deep surface drains:
 - The drains large enough to carry the flood water of the catchment area from the shallow surface drains and are of sufficient depths to provide outlets even for underground tile drains, if provided are known as deep surface drains.
 - These drains carry storm water discharge, drains, shallow surface drains and seepage water coming from underground tile drains.
 - Sub surface drains are required for soils with poor internal drainage as well as high water table.



6. Write short notes on:

- i. Surface drainage and its application
- ii. Subsurface drainage and its application
- iii. Drainage coefficient (D.C)

Answer:

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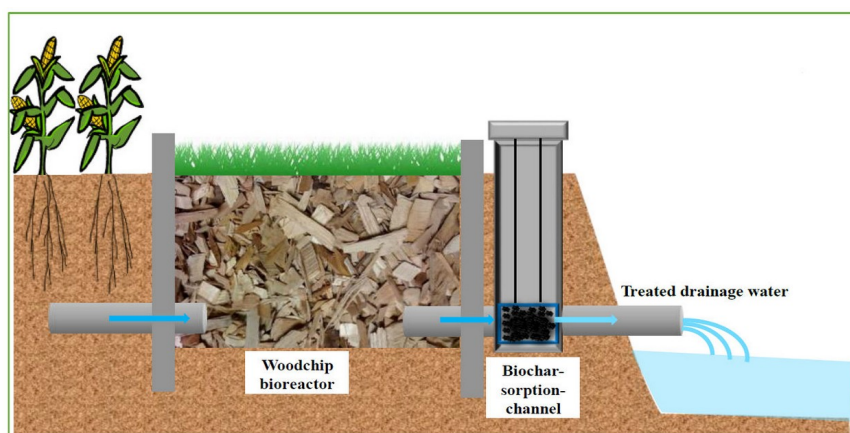
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Application of surface drainage:

- a. **Agricultural fields:**
Surface drainage removes excess rainwater and irrigation water to prevent waterlogging and improve crop growth.
- b. **Low-lying and flat areas:**
It drains stagnant water where natural drainage is poor.
- c. **Heavy clay soils:**
Surface drainage system removes surface water because infiltration is slow.
- d. **Flood-prone areas:**
The drainage helps safely convey floodwater to natural watercourses.

ii. Sub-surface drainage and its application:

- The removal of excess rainwater by using buried perforated pipes or tile drains to lower the groundwater table is known as subsurface drainage system.
- It is most effective for controlling water logging in agricultural fields.
- Sub-surface drains are required for high water table conditions in the areas where the soils are having poor internal drainage ability.
- If no impervious layer occurs below the farm land and the water table is low (lower than about 3 m from the ground), internal soil drainage may be sufficient and no tile drains are needed.
- For maximum productivity of most of the crops, both surface as well as sub-surface drains may sometimes become essential particularly in areas of higher water table.
- It consists of tile drains usually made up of porous earthen ware and circular in section with the diameter of 10 to 30 cm or so.
- These drains are laid below the ground level in the trenches which is backfilled with sand and excavated soil.



Applications of sub-surface drainage:

- a. **Agricultural lands:**
Sub-surface drainage system controls high groundwater levels and improves crop growth.
- b. **Waterlogged areas:**
It Lowers the water table and reclaims waterlogged land.

c. **Saline and alkali soils:**

It also removes excess salts from the root zone, improving soil fertility.

d. **Heavy clay soils:**

Sub-surface drainage also helps to improve soil aeration and root development.

e. **Irrigated areas:**

It prevents waterlogging caused by excessive irrigation.

iii. Drainage Coefficient:

Drainage Coefficient (DC) is the depth of excess water that must be removed from a drainage area within 24 hours or a specified time to prevent waterlogging. This ensures proper crop growth. It is usually expressed as: mm/day or cm/day.

$$\text{Drainage Coefficient} = \frac{\text{Depth of water to be drained}}{\text{Time}}$$

It is also used to determine the required drainage discharge:

$$Q = \frac{A \times DC}{86400}$$

where :

$$Q = \text{Drainage discharge (m}^3/\text{s)}$$

$$A = \text{Drainage area (m}^2\text{)}$$

$$DC = \text{Drainage coefficient (m / day)}$$

Factors affecting Drainage coefficient:

- Rainfall intensity and duration.
- Crop type and tolerance to waterlogging.
- Soil type and infiltration capacity.
- Topography and land slope.
- Irrigation practices.
- Groundwater table.

7. **What are the needs for controlling rivers in Nepal? List the summary of design procedure for river training works in hills and plains of Nepal.**

Answer:

In Nepal, the rivers of terai and inner terai plains are carried out with extensive river training works because of the high sedimentation and unpredictable flooding during the monsoon season. Koshi river which brings the second largest sediment deposition among all the rivers in the world is highly prone to frequent flooding and lateral shifting. Several projects are carried out to construct massive embankments and guide bunds to protect settlements and agricultural lands near Koshi river and its tributaries.

Hence, River control is essential in Nepal due to the presence of steep rivers in hilly region, large amounts of sediments in terai region, and high flows during the monsoon. Proper river control helps protect lives, property, and infrastructure while supporting sustainable development.



The needs for controlling rivers in Nepal are listed below:

- i. Reduces flood damage to lives, property, and agricultural land.
- ii. Prevents loss of fertile land and protects settlements.
- iii. Safeguards bridges, roads, irrigation canals, hydropower projects, and other structures.
- iv. Prevents excessive shifting and meandering of river courses.
- v. Controls excessive sediment deposition that reduces river and reservoir capacity.
- vi. Prevents crop damage and loss of productive farmland.
- vii. Helps maintain river ecology and minimizes land degradation

The design procedure for the river training works in hills and plains of Nepal are summarized below:

- i. Preliminary investigation:
 - The hydrological, hydraulic, topographical, and geological data are collected.
 - River morphology, sediment transport, and flood history are also properly studied and analyzed.
 - Field surveys are conducted and problem areas are identified and analyzed.
- ii. Design flood estimation:
 - The design discharge is determined using suitable hydrological methods.
 - Appropriate return period is selected depending on the importance of the structure.
- iii. Hydraulic analysis:
 - Flow velocity, water level, scour depth, and afflux are calculated.
 - The channel stability and sediment movement are also analyzed.
- iv. River training alignment:
 - In hilly region, the existing stable river course must be followed with minimum interference.
 - In plain lands, meandering rivers are stabilized and the river alignment is fixed using river training works.
- v. Selection of river training works:
 - In hilly region, check dams, spurs, retaining walls, revetments, gabions, and bio-engineering measures are provided as river training measures.
 - Similarly, in plain regions like terai embankments (levees), guide banks, spurs (groynes), revetments, launching aprons, and floodwalls.
- vi. Structural design:
 - Embankments, spurs, guide banks, revetments, and aprons are designed.
 - The dimensions, side slopes, freeboard, stone size, and filter layers are determined.
 - The foundations are designed considering expected scour depth.

vii. Protection works:

- Protection works such as stone pitching, gabion mattresses, riprap, vegetative protection, and launching aprons are constructed where required.

viii. Environmental and social consideration:

- Environmental impacts must be minimized.
- The aquatic habitats and nearby communities must be protected from natural calamities.
- Bio-engineering and watershed conservation measures must be adopted.

ix. Construction and quality control:

- The suitable construction materials and methods are used.
- Proper supervision and quality assurance must also be ensured.

x. Operation and maintenance:

- Regular inspection after floods.
- Repair of damaged structures.
- Removal of deposited sediments and maintain river training works.

8. **What is runoff? What are the factors that affect runoff from a catchment area? Describe the hydrological cycle in brief with neat sketch.**

Answer:

The flow of rainwater where it travels all over the surface as overland flow and through the channel as open channel flow and reaches the catchment outlet is called runoff. The runoff can be classified as:

i. Direct runoff:

It is that part of runoff that enters the stream immediately after the rainfall.

ii. Base flow:

The delayed flow that reaches a stream or ground water flow is known as baseflow.

The factors affecting runoff are listed below:

i. Type of precipitation:

- Runoff depends upon the type of precipitation.
- Increased amount of rainfall contributes to a greater amount of runoff in water bodies as well as ground surface if the runoff coefficient of the area is high.

ii. Duration of rainfall:

- When the duration of rainfall increases, infiltration capacity of the ground decreases resulting more runoff.

iii. Size of basin:

- Larger the watershed, longer time it requires to deliver runoff to the outlet.

iv. Shape of watershed:

- In fan shaped watershed, Runoff from the nearest tributaries drained out before the floods of farthest tributaries, peak runoff is less.
- In broad shaped watershed, all tributaries contribute runoff almost at the same time so greater peak flow occurs.

v. Land use:

- More forest area reduces the runoff due to increased infiltration and percolation.

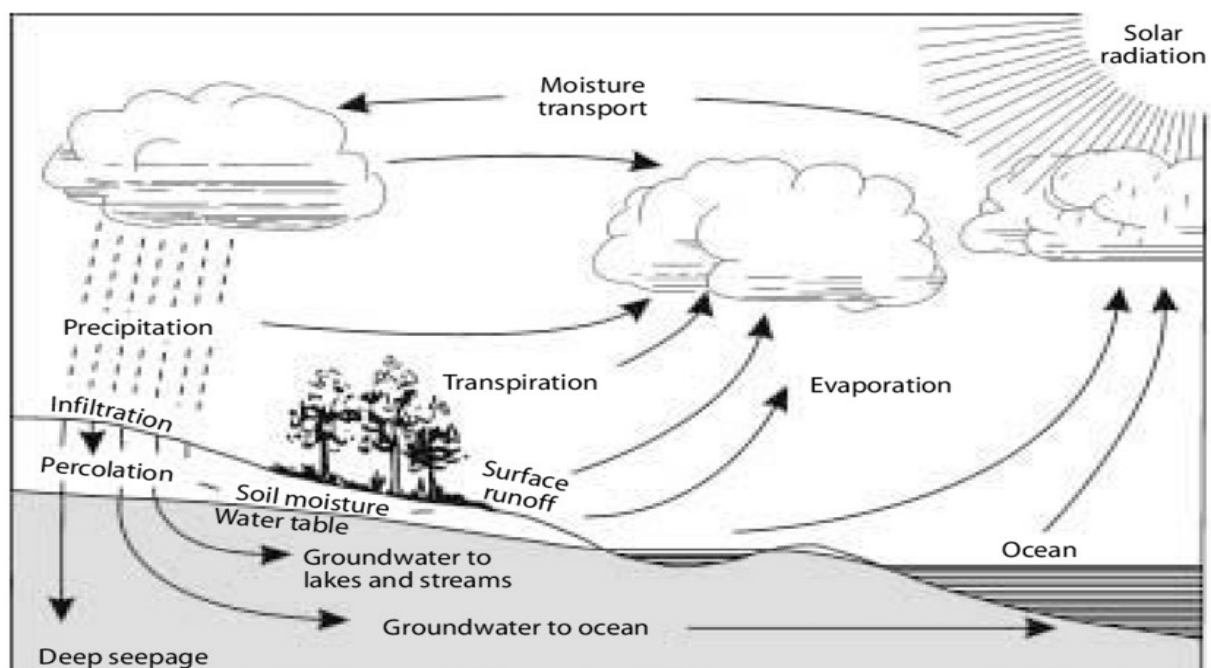
- In the meantime, in urban areas, runoff is more due to less infiltration and percolation and absence of groundwater recharge.
- The terrible problem of surface runoff is being experienced by the residents of Kathmandu valley during monsoon season, the main reason of it being increased built-up area and less open spaces and forest area.

vi. Topographic characteristics:

- Higher the slope, faster the runoff.

Hydrological cycle:

- Hydrological cycle is also known as water cycle.
- It is a cycle that involves continuous circulation of water in earth's atmosphere system.
- The cycle has no beginning or end but its process occurs continuously.
- The cycle involves exchange of energy that leads to temperature change.
- The schematic diagram of hydrological cycle is given below:



- The evaporation phase of hydrological cycle purifies the water and replenish the land with fresh water.
- The water from oceans, river, stream, lakes evaporate, move upward and forms clouds by absorbing energy from the sun.
- Water falls back to the sources in the form of rain and also get shifted towards the land due to wind.
- Some portion of rainfall falling on land may get evaporated into the atmosphere while falling.
- Some portion of rainfall may be intercepted by vegetation, leaves, forest floor which later gets evaporated or fall onto the ground.
- Some portion of precipitation falls directly on the ground surface and gets infiltrated into the soil while the other portion runs above and below the ground.
- The infiltrated water may run deep into the crevices and earth reaches to be stored as aquifer by deep percolation.
- Also, the absorbed water by vegetation returns to the atmosphere by transpiration.
- There are two main components of hydrological cycle:
 - Transportational components: Precipitation, Evaporation, Transpiration, Infiltration and Runoff
 - Storage components: Soil moisture, ground water storage, storage on land.

9. Given below are observed flows from a storm of 6 hours duration on a stream with a catchment area of 500 km². Assumes the base flow to be zero, derive the ordinates of a 6-hour unit hydrograph.

Time hours	0	6	12	18	24	30	36	42	48	54	60	66	72
Observed flow m ³ /s	0	100	250	200	150	100	70	50	35	25	15	5	0

Answer:

Given,

Catchment area, A=500 km²=500×10⁶ m²

Storm duration = 6 hours

Base flow = 0

Using the trapezoidal rule with a time interval of 6 hours = 21600 s:

Interval	Average Flow (m ³ /s)	Volume (m ³)
0–6	50	1,080,000
6–12	175	3,780,000
12–18	225	4,860,000
18–24	175	3,780,000
24–30	125	2,700,000
30–36	85	1,836,000
36–42	60	1,296,000
42–48	42.5	918,000
48–54	30	648,000
54–60	20	432,000
60–66	10	216,000
66–72	2.5	54,000

Total Direct Runoff Volume, V= 21,600,000 m³

Runoff depth: $d = \frac{V}{A} = \frac{21.6 \times 10^6}{500 \times 10^6} = 0.0432 \text{ m} = 4.32 \text{ cm}$

Thus, the storm produced 4.32 cm of direct runoff.

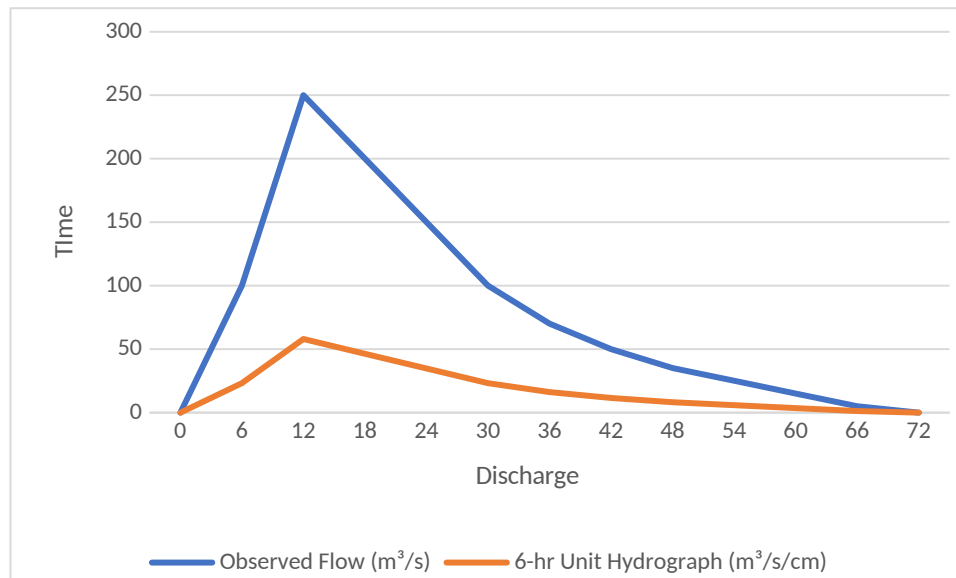
Deriving 6 hours unit hydrograph ordinates:

A Unit Hydrograph corresponds to 1 cm of runoff.

$$\text{UH Ordinates} = \frac{\text{Observed Flow}}{4.32}$$

Time (hr)	Observed Flow (m ³ /s)	6-hr Unit Hydrograph (m ³ /s/cm)
0	0	0.00
6	100	23.15
12	250	57.87
18	200	46.30
24	150	34.72
30	100	23.15

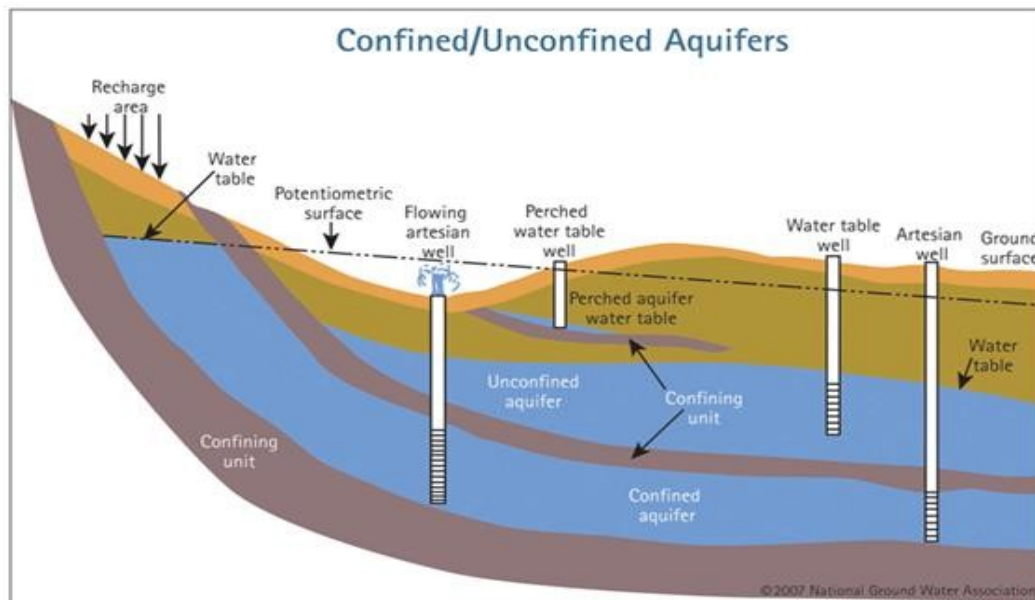
Time (hr)	Observed Flow (m ³ /s)	6-hr Unit Hydrograph (m ³ /s/cm)
36	70	16.20
42	50	11.57
48	35	8.10
54	25	5.79
60	15	3.47
66	5	1.16
72	0	0.00



10. What is aquifer? Describe the different types of wells and their classification. Describe the different types of wells for different types of aquifers.

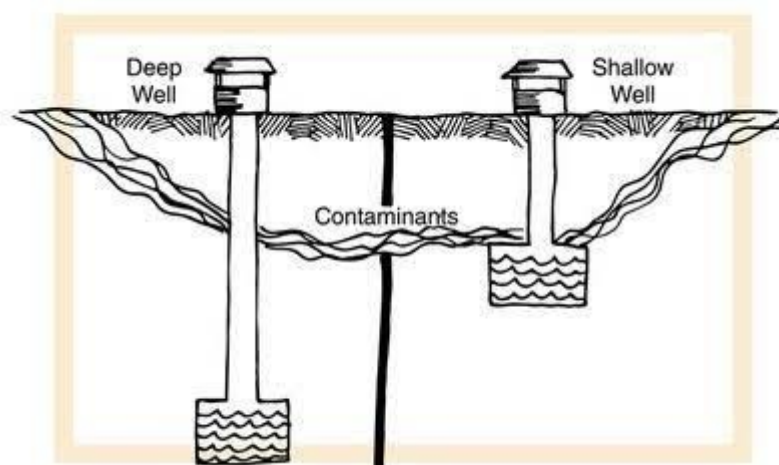
Answer:

- An aquifer may be defined as the geological formation that contains sufficient permeable material that permits storage as well as transmission of water through it under ordinary conditions.
- Water can be withdrawn from aquifer for example: sand layer.
- Aquifer are of two types:
 - i. Confined aquifer:
 - It is the type of aquifer in which ground water is confined under pressure greater than atmospheric pressure by overlying relatively impermeable strata.
 - Confined aquifer generally occurs at significant depth below the ground surface.
 - It is known as artesian aquifer or pressure aquifer.
 - ii. Unconfined aquifer:
 - It is the type of aquifer in which water table forms the upper surface of zone of saturation.
 - It is also known as phreatic aquifer or non-artesian aquifer.



Wells:

- Wells are the vertical holes made below the ground surface to tap water.
- The maximum quantity of water that can be guaranteed during lean period is known as the safe yield of the well.
- Water wells may be classified as:
 - i. Open wells or dug wells:
 - They are comparatively of larger diameter and larger depth but low yields.
 - The diameter ranges from 1 to 10 m usually with the depth at the range of 2 to 20 m.
 - The thickness of open wells varies from 50 cm to 75 cm depending on the depth of the well.
 - Open wells may further divided into:
 - a. Shallow open wells:
 - ✓ Shallow open wells are those which rest in the top water-bearing strata and draw their supplies from the surroundings.
 - ✓ There is always a risk of contamination because water is obtained from the top most pervious layer.
 - b. Deep open wells:
 - ✓ Deep open wells are those which rest on impervious strata and draw their supplies from pervious formation lying below the impervious strata.
 - ✓ The quantity of water is greater in this well than in shallow wells.
 - ✓ Deep well water has better quality but sometimes it may contain dissolved impurities as it depends upon geographical conditions.



ii. Tube wells:

- Tube well is a long pipe sunk into the ground intercepting one or more water bearing strata.
- As compared to open wells the diameter of tube wells is much less.
- The depth of tube wells up to 30 m is shallow tube wells and up to 600 m are deep tube wells.
- The tube wells may be classified as:
 - a. Strainer type tube wells:
 - It is the most common and widely used tube well.
 - The pipe introduced into the ground is an assembly of strainer pipes and blind pipes that are alternately placed.
 - The length of strainer and blind pipes are so adjusted that the strainer pipe rest against the water bearing strata and blind pipes rest against impervious strata.
 - b. Cavity type tube wells:
 - A cavity-type tube wells consists of a pipe sunk into the ground and resting on the bottom of strong strata.
 - The initial stage of pumping fine sand comes out with water consequently, a cavity is formed at the bottom.
 - c. Slotted type tube wells:
 - It consists of a pipe that is slotted for part of its length at one end and the rest of the length is a plain pipe.
 - The mixture of gravel and coarse sand is placed around the pipe to prevent the fine sand from entering the well pipe.
 - The slotted portion of the pipe is surrounded by a mixture of gravel and coarse sand called shrouding which is filled between casing pipe and slotted pipe during withdraw of casing.
 - d. Perforated type tube wells:
 - It is a shallow depth tube well and is used for a short duration such as in construction site.
 - Pipes having covered by jute ropes act as a strainer and prevents the fine particles from entering the well pipe.

The different types of wells suitable for different types of aquifers are listed below:

i. Open well:

- Suitable for unconfined (shallow) aquifers.
- Large diameter (1–5 m) and shallow depth.

ii. Tube well:

- Suitable for confined and unconfined aquifers.
- Small diameter and deep.
- Constructed by drilling and fitted with casing and screens.

iii. Artesian well:

- Suitable for confined aquifers.
- Water rises naturally due to artesian pressure.

iv. Driven well:

- Suitable for shallow unconfined aquifers in sandy or gravelly soils.
- Constructed by driving a small-diameter pipe into the ground.

v. Bored well:

- Suitable for soft formations and shallow unconfined aquifers.
- Constructed using hand or power augers.

vi. Radial well:

- Suitable for alluvial aquifers near rivers.
- Consists of a central well with horizontal perforated pipes extending into the aquifer.